

Tirth Gada

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Education

Veermata Jijabai Technological Institute (VJTI) , Mumbai, India	2024 – 2028
Second Year B.Tech in Electrical Engineering, Minor in Robotics	CGPA: 8.78
Champions Science Junior College , Mumbai, India	2022 – 2024
HSC (12th Std)	Percentage: 83.13%

Experience

Research Intern , HiRO Lab , Indian Institute of Science (IISc) , Bangalore	May 2026
<i>Multi-Embodiment Grasping</i>	
- Pretraining multi-embodiment grasping models such as kinematics-flow and geomatch for generalising success rates across multiple grippers, with the best grasp model across 5 grippers (Panda, Shadow, Allegro, Dexee, VX300) reaching 86.1% mean success rate by epoch 120. - Worked on extending semantic understanding grasping frameworks such as GraspMAS and ShapeGrasp for task-agnostic 6-DoF grasp and task planning.	

Projects

Robot Learning	July 2026
<i>Sim2Real Transfer · Imitation Learning · ManiSkill3 · LeRobot</i>	
- Implemented Flow Policy Optimization (FPO) for ManiSkill3-based visual manipulation of the SO100 robotic arm, and benchmarked PPO vs. FPO performance in both simulation and real-world deployment. - Trained ACT and fine-tuned SmolVLA vision-language-action models with visuotactile feedback from FlexiTac sensors, achieving robust imitation learning for contact-rich tabletop manipulation tasks on the SO100. - Developed Karma, a config-driven realtime policy runtime for bimanual YAM robots, and deployed B-spline diffusion policies for smooth, jerk-limited action generation during manipulation. - Published residual RL work on frozen VLA fine-tuning at the RL4VLA Workshop, RSS 2026.	
Open Duck Mini : Bipedal Locomotion	December 2025
<i>Sim2Real · Reinforcement Learning · Raspberry Pi</i>	
- Built an open-source 3D-printed bipedal robot platform designed for accessible and low-cost locomotion research. - Integrated ONNX-based RL policy inference on Raspberry Pi for real-time onboard control, establishing a complete sim-to-real transfer pipeline targeting stable bipedal walking behaviors.	

Research

Hill-Climbing a Real-World VLA in Simulation: Frozen-Backbone Residual RL for MolmoAct2	
Atharva Kshirsagar, Tirth Gada , Vrushtee Gaikwad	<i>RL4VLA Workshop, RSS 2026</i>
Proposed freezing a deployed VLA (MolmoAct2) and learning a bounded residual policy via TD3 in ManiSkill simulation, recovering a miscalibrated 14-DoF bimanual setup without updating the billion-parameter backbone.	

Competitions

Smart India Hackathon 2025 – Grand Finale , IIT Kharagpur	December 2025
<i>Finalist (Selected from 10,000+ teams, Problem Statement 25056)</i>	
- Developed a novel electrochemical sensor system using Linear Sweep Voltammetry (LSV) for real-time gold detection in mining environments. - Designed a multi-stage signal processing pipeline incorporating baseline correction, Savitzky-Golay smoothing, and derivative-based peak detection for robust mineral identification from noisy voltammetric signals.	

Technical Skills

Programming Languages: C, C++, Python
Software Frameworks: ManiSkill3, LeRobot, ESP-IDF, PyTorch
Hardware Platforms: ESP32, Raspberry Pi, Motor Controllers, Sensors
Development Tools: Git, VS Code, Linux, Docker

Extracurriculars

Society of Robotics and Automation (SRA) , VJTI, Mumbai	July 2025 – Present
<i>Software Head</i>	
- Conducted the Wall-E workshop for 200+ first-year students on embedded C, PID control, sensor integration (MPU6050, LSA), and ESP32 firmware with ESP-IDF/FreeRTOS; also competed in the SRA Autosim challenge, implementing a maze-solving robot on the legacy SRA ESP32 board. - Conducted the Pixels seminar on computer vision fundamentals in C++, covering convolution, morphological operations, blob detection, and OpenCV. - Conducted the MARIO workshop on a 3-DOF robotic manipulator with ROS2 and micro-ROS, covering DH parameters, forward/inverse kinematics, Gazebo, Rviz and Mujoco simulation, and ESP32 servo control.	